

Mini Project

AIM : Syntax Analyzer for if-else Statements

Pseudo code :

BEGIN

PRINT "Enter Python if-else statement:"

READ input_code

// Check basic structure

IF input_code does NOT contain "if" AND ":"

PRINT "Syntax Error: Missing 'if' or colon"

EXIT

// Check indentation

IF indentation is inconsistent (mix of tabs/spaces)

PRINT "Syntax Error: Inconsistent indentation"

EXIT

// Check for elif/else blocks

IF contains "elif"

IF "elif" is not aligned with "if" OR missing ":"

PRINT "Syntax Error: Invalid elif block"

EXIT

IF contains "else"

IF "else" is not aligned with "if" OR missing ":"

PRINT "Syntax Error: Invalid else block"

EXIT

```
// Check condition parentheses (optional in Python)
IF "if" followed by "(" but no matching ")"
    PRINT "Syntax Warning: Unclosed parentheses"

PRINT "Valid if-else syntax!"

END
```

Code :

```
def check_indentation(code_lines):
    """Check for consistent indentation (no mixing tabs and spaces)"""
    for line in code_lines:
        if line.strip(): # Skip empty lines
            if '\t' in line and ' ' in line[:line.find(line.lstrip()[0])]:
                return False
    return True

def check_colon(line):
    """Verify colon exists after if/elif/else"""
    return ':' in line.split('#')[0] # Ignore comments

def check_parentheses(line):
    """Check balanced parentheses in condition"""
    condition = line.split(':')[0]
    return condition.count('(') == condition.count(')')

def analyze_if_else(lines):
    has_if = False
    base_indent = -1
```

```
for i, line in enumerate(lines, 1):  
    stripped = line.strip()  
  
    # Skip empty lines and comments  
    if not stripped or stripped.startswith('#'):  
        continue  
  
    # Check if line  
    if stripped.startswith('if'):  
        has_if = True  
        base_indent = len(line) - len(line.lstrip())  
  
        if not check_colon(line):  
            print(f"Line {i}: Syntax Error → Missing colon after 'if'")  
            return  
  
        if not check_parentheses(line):  
            print(f"Line {i}: Syntax Warning → Unbalanced parentheses in 'if' condition")  
  
    # Check elif  
    elif stripped.startswith('elif'):  
        if not has_if:  
            print(f"Line {i}: Syntax Error → 'elif' without preceding 'if'")  
            return  
  
    current_indent = len(line) - len(line.lstrip())  
    if current_indent != base_indent:
```

```
print(f"Line {i}: Syntax Error → 'elif' not aligned with 'if'")  
return
```

```
if not check_colon(line):  
    print(f"Line {i}: Syntax Error → Missing colon after 'elif'")  
    return
```

```
# Check else
```

```
elif stripped.startswith('else'):  
    if not has_if:  
        print(f"Line {i}: Syntax Error → 'else' without preceding 'if'")  
        return
```

```
current_indent = len(line) - len(line.lstrip())  
if current_indent != base_indent:  
    print(f"Line {i}: Syntax Error → 'else' not aligned with 'if'")  
    return
```

```
if not check_colon(line):  
    print(f"Line {i}: Syntax Error → Missing colon after 'else'")  
    return
```

```
if not has_if:  
    print("Syntax Error → No 'if' statement found")  
    return
```

```
print("✅ Valid if-else syntax!")
```

```
def main():  
    print("Python if-else Syntax Analyzer")  
    print("Enter your code (blank line to finish):")  
  
    lines = []  
    while True:  
        try:  
            line = input()  
            if line == "" and lines and lines[-1] == "":  
                break  
            lines.append(line)  
        except EOFError:  
            break  
  
    # Remove trailing empty lines  
    while lines and lines[-1] == "":  
        lines.pop()  
  
    if not lines:  
        print("Error: No input provided")  
        return  
  
    # Check indentation first  
    if not check_indentation(lines):  
        print("Syntax Error → Mixed tabs and spaces in indentation")  
        return  
  
    analyze_if_else(lines)
```

```
if __name__ == "__main__":  
    main()
```

Output :

```
PS D:\mini-project> python -u "d:\mini-project\if_else_checker.py"  
Python if-else Syntax Analyzer  
Enter your code (blank line to finish):  
if x > 5:  
    print("Greater")  
elif x == 5:  
    print("Equal")  
else:  
    print("Smaller")  
  
✔ Valid if-else syntax!  
PS D:\mini-project>
```

```
PS D:\mini-project> python -u "d:\mini-project\if_else_checker.py"  
Python if-else Syntax Analyzer  
Enter your code (blank line to finish):  
if x > 5:  
    print("Greater")  
elif x == 5:  
    print("Equal")  
else:  
else  
  
Line 6: Syntax Error → Missing colon after 'else'  
PS D:\mini-project>
```

Flow Chart :